Marine microbiology
Lecture 2 - 3
Diversity of Microorganisms

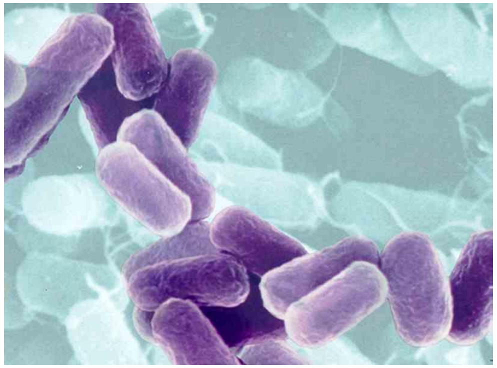
Objectives for this lecture:

- At the end of this lecture, you should be able to:
  - Explain taxonomy（分类学）and taxonomic ranks
  - Name the three domains of life and explain how they were determined
  - List the differences between prokaryotes(原核生物) and eukaryotes(真核生物)
  - Compare archaea to eubacteria and eukaryotes
  - List some characteristics of: fungi; protists(原生生物)：slime moulds（粘菌类）, algae and protozoa(无脊椎原生动物)
  - Describe the lifestyle of viruses and explain why they do not appear in the Universal phylogenetic tree(进化树)

Taxonomy: What's in a name?

- Taxonomy is:
  - Naming and grouping of organisms
- Taxonomy involves:
  - Classification
    - Arrangement of organisms into groups (taxa) based upon common features
  - Nomenclature
    - Assignment of names according to established rules
  - Identification
    - Using distinguishing features to detect taxa

Balantidium coli（肠袋虫属 ）



Escherichia coli（埃希氏杆菌属）

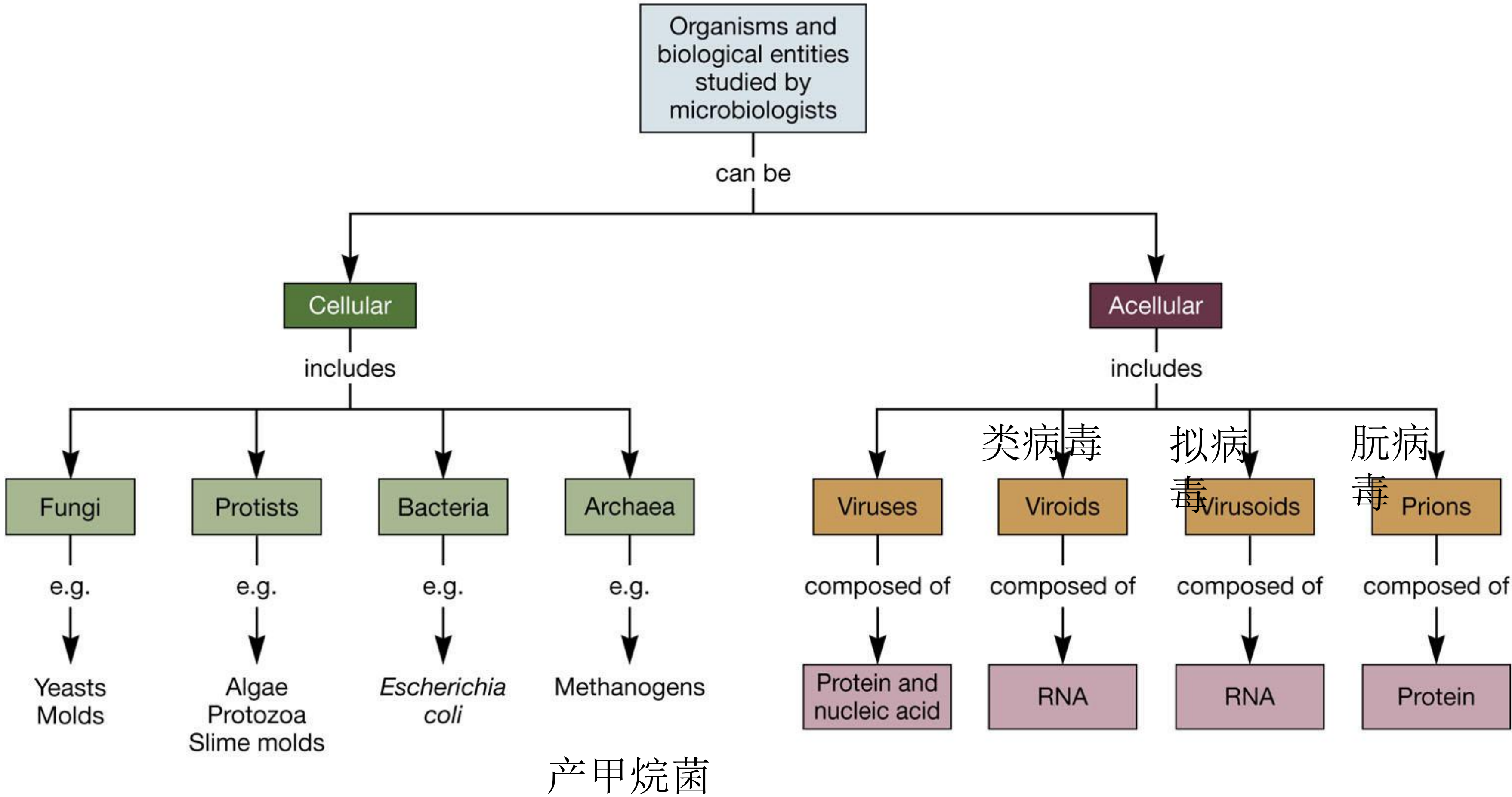
Taxonomic ranks

- 1735: Carl von Linnaeus
- Microbes are named using binomial system of Linnaeus
  - Genus
    - capitalized, italicized (e.g. *Homo*)
  - Species
    - lower case, italicized (e.g. *sapiens*)
- Hierarchical（分层的） classification: each rank shares a common set of features
  - Kingdom（界）
  - Phylum（门）
  - Class（纲）
  - Order（目）
  - Family（科）
  - **Genus（属）**
  - **Species（种）**

An example of taxonomic ranks in bacteria

Domain	***Bacteria***
Phylum	***Proteobacteria*** 变形菌门
Class	***gamma-Proteobacteria***
Order	***Enterobacteriales*** 肠道菌
Family	***Enterobacteriaceae***
Genus	***Shigella*** 志贺氏杆菌
Species	***S. dysenteriae*** 痢疾菌

Diversity of Microbes



Diversity of Microbes

Naked eye

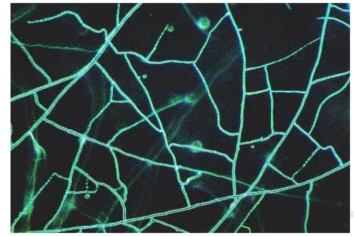
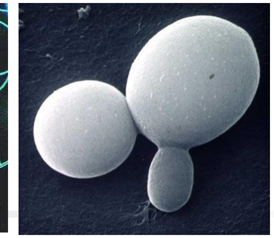
Light microscope

Electron microscope

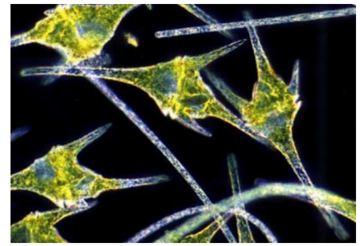
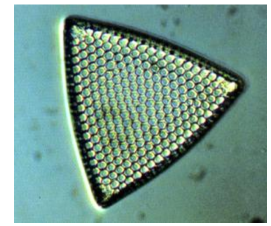
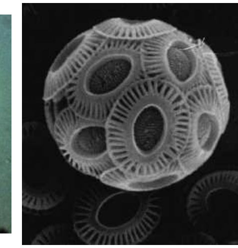
Decreasing size

- **Range:**
  - **eukaryotic microbes**
    - Fungi
    - Microalgae
    - Protozoans 无脊椎原生动物
  - **prokaryotic microbes** ("bacteria")
    - Eubacteria
    - Archaea
  - **viruses**

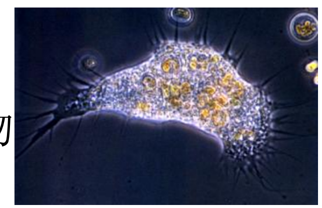
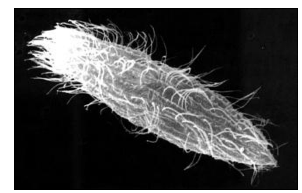
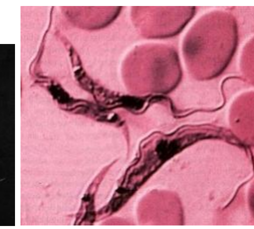
Fungi


Microalgae

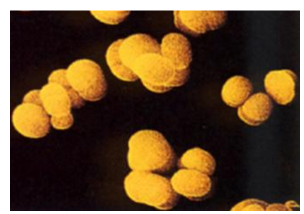




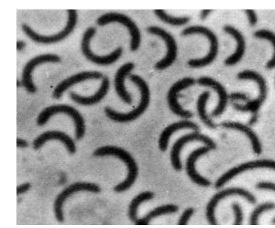
Protozoans
无脊椎原生动物

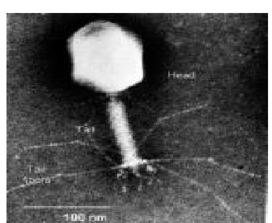
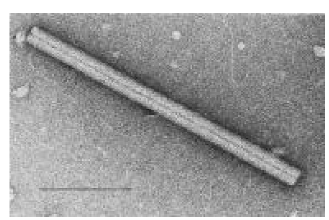
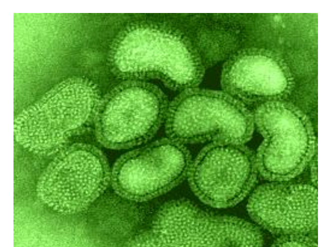




Eubacteria

Archaea

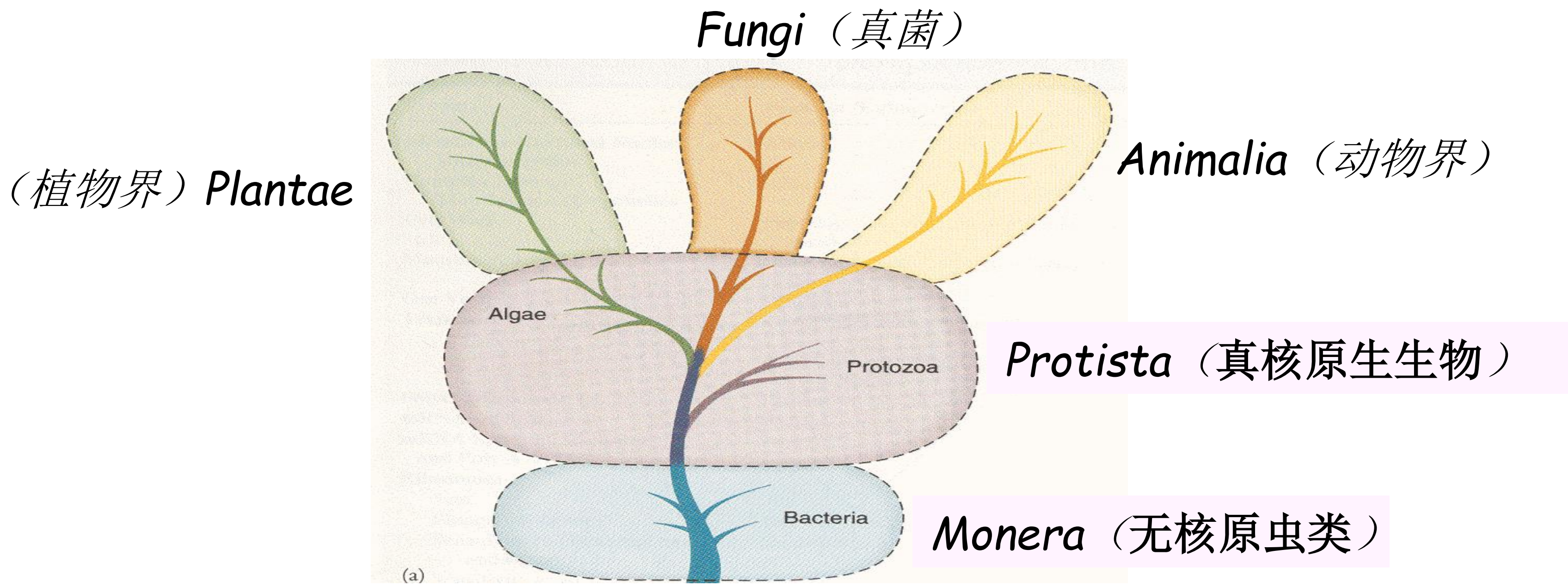




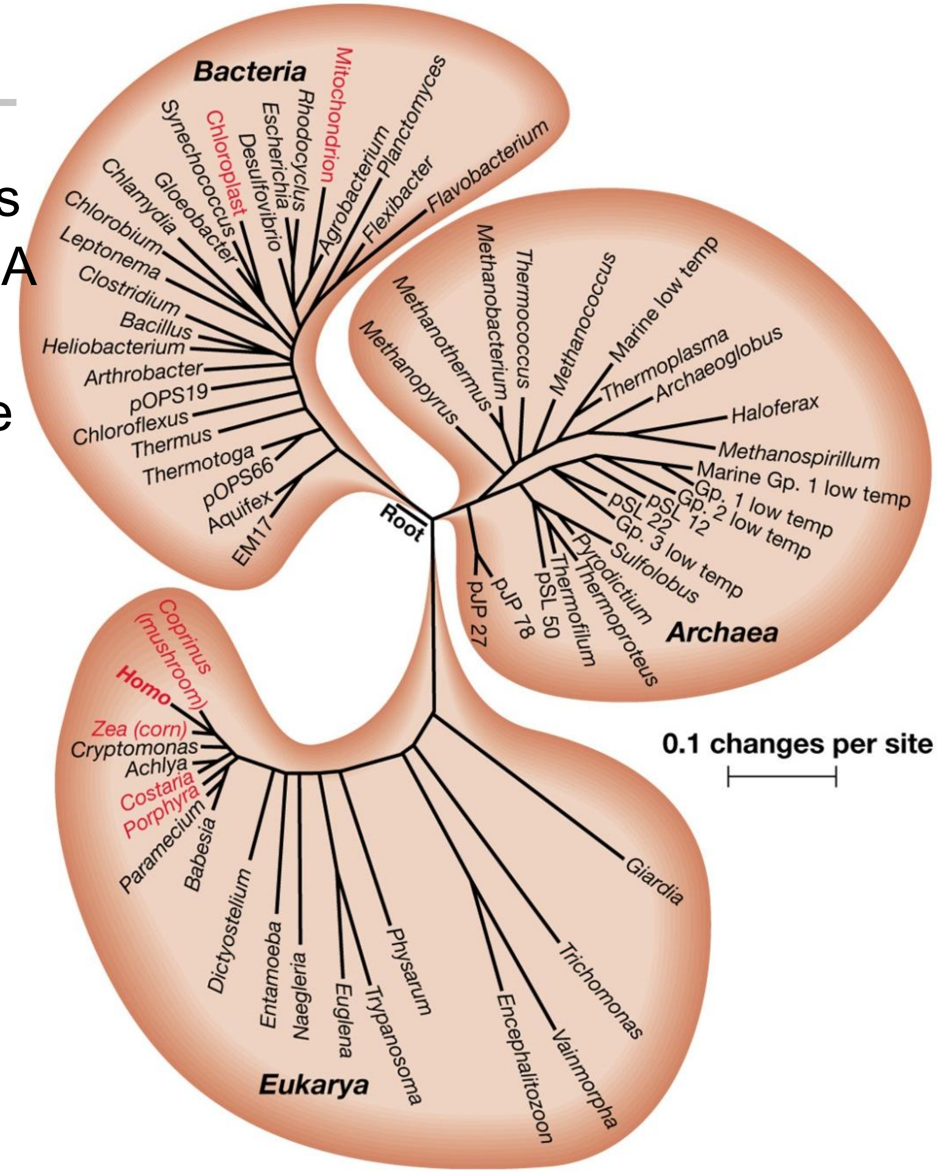
Domains/Kingdoms of life

- Previous classification scheme divided all organisms into five kingdoms
- Microbes were placed in three of these kingdoms
- This scheme is <u>no longer accepted</u> by microbiologists



The three domains of life

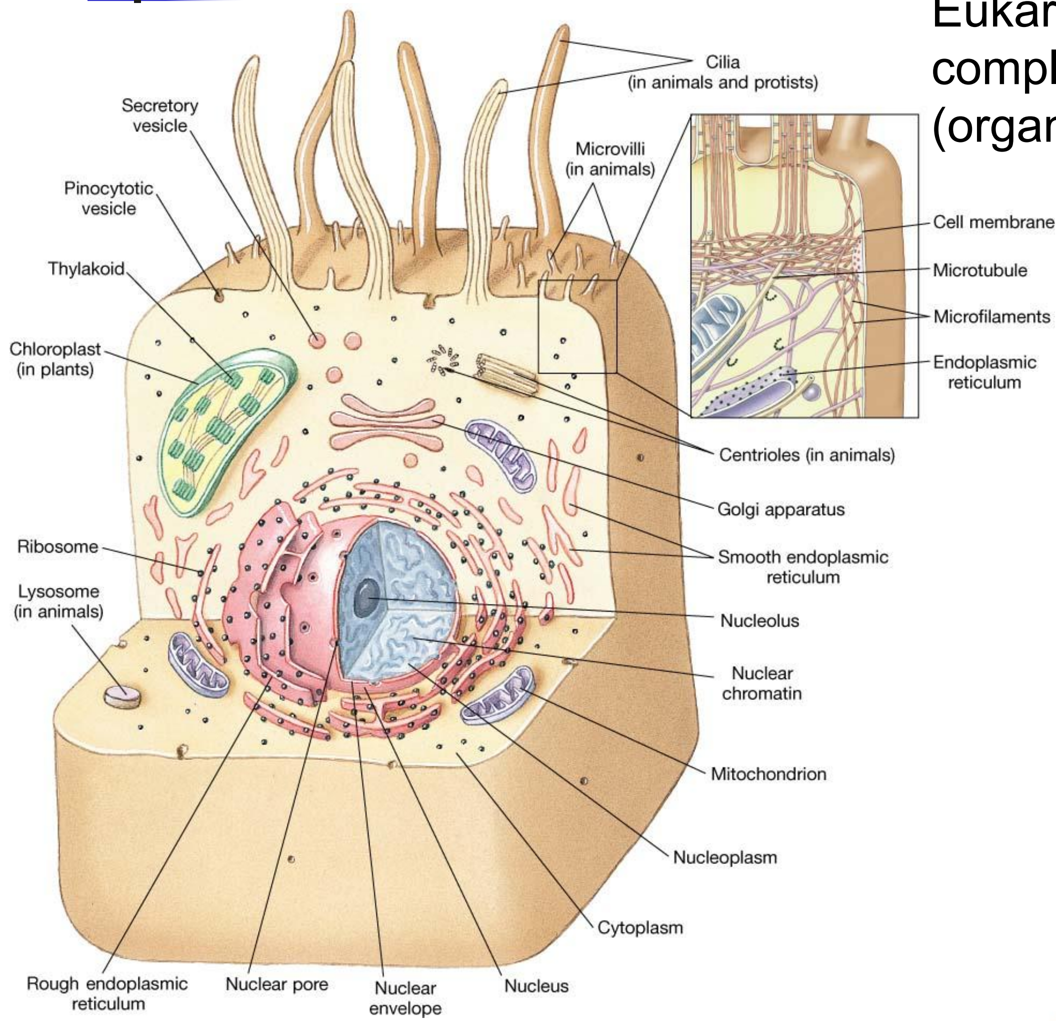
- In the 1970s, Carl Woese and others pioneered the use of ribosomal RNA (rRNA) sequencing
- rRNA sequence provides a measure of evolutionary relatedness
- Three distinct lineages（进化分支）of cellular life were identified, called domains
  - **Bacteria (Eubacteria)**
  - **Archaea**
  - **Eukarya**
- Both Bacteria and Archaea are prokaryotes
- All Eukarya are eukaryotes



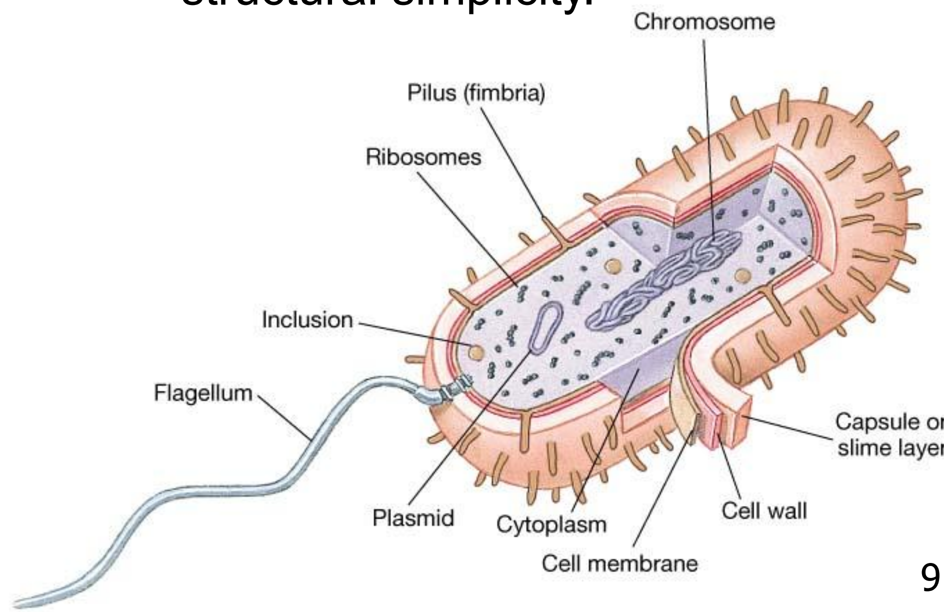
Universal phylogenetic tree: shows evolutionary relationships based on rRNA sequence comparisons

Prokaryotes and Eukaryotes

Eukaryotic cell showing relative structural complexity. Note compartmentalization (organelles, nucleus)



Prokaryotic cell showing relative structural simplicity.

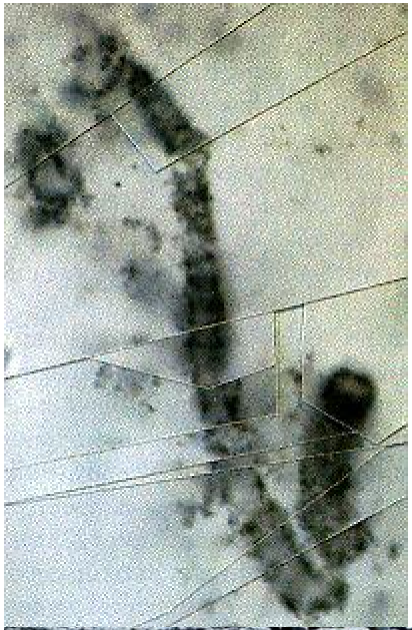


Comparison of Prokaryotes and Eukaryotes

Characteristics	Prokaryotic cell	Eukaryotic cell
Size of cell	Usually smaller (1-100 uM^2)	Usually larger (1-1000,000 uM^2)
Cell wall	Contains peptidoglycan（肽聚糖）	No peptidoglycan present
Genetic material	Circular DNA molecule. No discrete （离散的） nucleus	Arranged in chromosomes. Nucleus present in cell
Mitosis and meiosis	Absent	Present
Ribosomes	Smaller in size than eukaryotes, 70S. Free in cytoplasm	Larger in size. 80S. On endoplasmic reticulum(内质网)
Membrane-enclosed organelles	Absent	include mitochondria, chloroplasts, Golgi complexes, lysosomes（溶酶体）, endoplasmic reticulum
Plasma membrane	Sterols（甾醇类；固醇类） usually absent	Sterols usually present
Site of respiration	Plasma membrane	Mitochondria(线粒体)
Site of photosynthesis	Internal membranes	Chloroplasts（叶绿体）
Locomotion	Flagella（鞭毛） that rotate, of simple composition. Some glide（滑行）	Flagella and cilia that undulate. Amoeboid movements

Microbial Evolution and Diversity

- Fossilized（石化的）prokaryote
  - 3.5-3.8 billion years old
  - Isolated in fossilized stromatolites（叠层石）
  - Domed（半球形的）, layered rocks formed by microbes and mineral sediments（沉积物）
- Earliest prokaryotes were anaerobes (do not require oxygen for growth)
- Early evolutionary division of prokaryotes into Bacteria and Archaea



Fossilized prokaryote

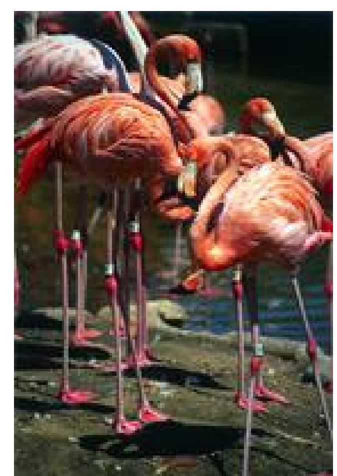
Stromatolites, Shark Bay, Western Australia

Comparison of Bacteria, Archaea and Eukaryotes

Characteristic	Bacteria	Archaea	Eukaryotes
Cell wall	***Contains muramic（胞壁酸） acid***	Lacks muramic acid	Lacks muramic acid
Membrane lipids	Ester-linked straight hydrocarbon chains	***Ether-linked branched aliphatic（脂质的） hydrocarbon chains***	Ester-linked straight hydrocarbon chains
Membrane-bound nucleus	Absent	Absent	***Present***
Chromosome	Single circular chromosome	Single circular chromosome	***Linear chromosomes***

Archaea (*Halobacterium*) in a salt pond➔red pigment（色素）

Halobacterium sp



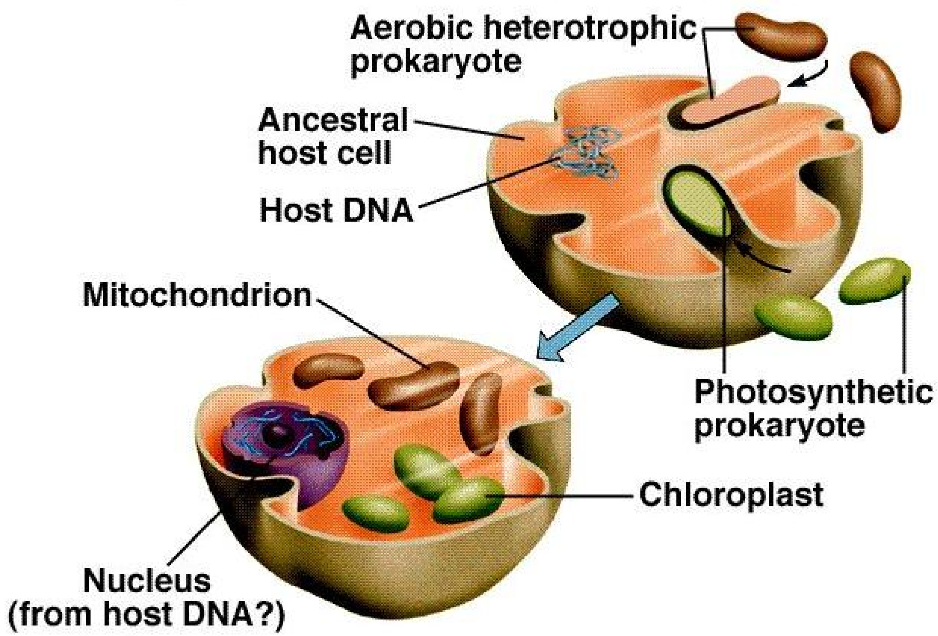
The pink flamingo（火烈鸟）feed in saline lakes containing *Halobacterium*

Comparison of Bacteria, Archaea and Eukaryotes

Characteristic	Bacteria	Archaea	Eukaryotes
RNA polymerase	***One type, with 6 different subunits***	Several types, each with 8 to 12 different subunits	Several types, each with 12 to 14 subunits
Ribosomes	70S	70S	***80S (70S in mitochondria and chloroplasts)***
Amino acid carried by initiator tRNA	***Formylmethionine（甲酰甲硫氨酸）***	Methionine	Methionine
tRNA	Thymine（胸腺嘧啶）and dihydrouracil usually present	***Thymine absent and dihydrouracil（二氢尿嘧啶）usually absent***	Thymine and dihydrourcil usually present
Cellular organelles	Absent	Absent	***Present***
Sensitivity to chloramphenicol & kanamycin antibiotics	***Sensitive***	Resistant	Resistant

Evolution of Microbes: Prokaryote to Eukaryote

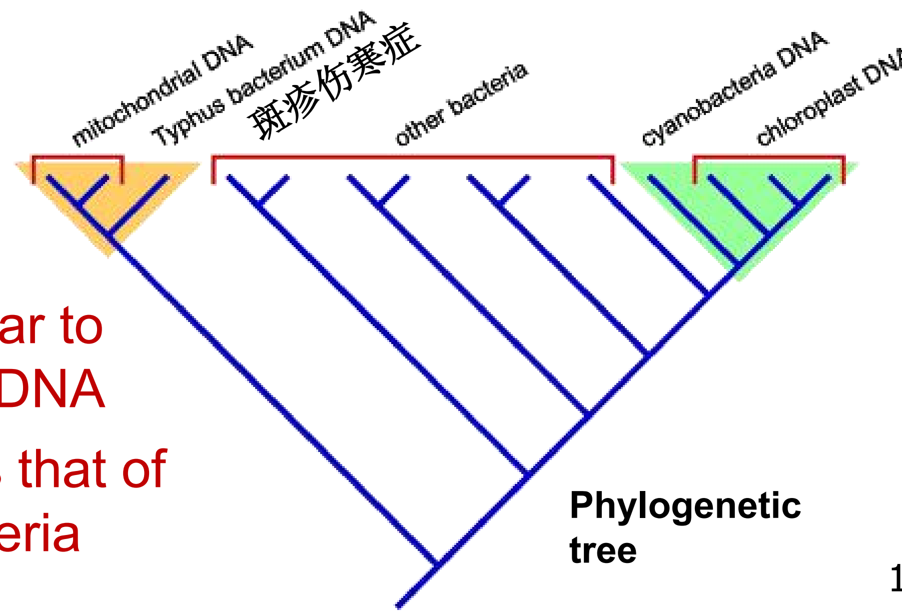
- Some observations on mitochondria and chloroplasts
  - Similar size to bacteria
  - Have their own genomes
    - Circular, and order of the genes are similar to a prokaryote
  - Protein translation
    - Use formylmethionine
  - Antibiotics that inhibit bacteria also inhibit mitochondria and chloroplasts but have no effect on eukaryote cell
- Leads to Endosymbiotic Hypothesis(内共生假说)



Model of endosymbiotic evolution of eukarya

Endosymbiotic Hypothesis

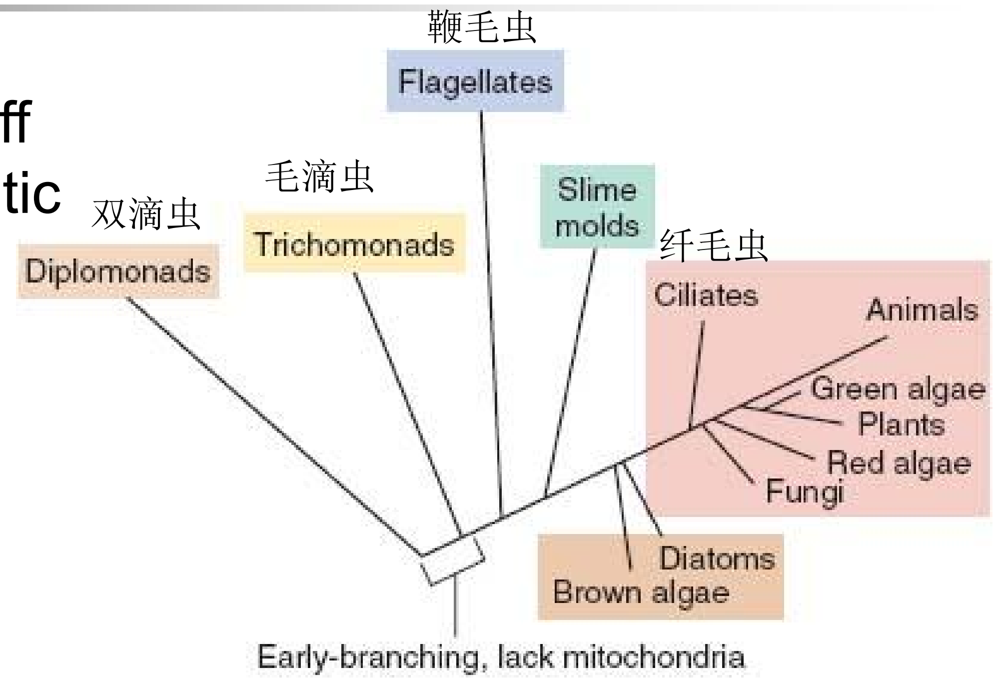
- Endosymbiosis is an interaction between two organisms in which one organism lives inside the other
- Over time, bacterial endosymbionts of an ancestral（祖先的）cell(s) lost their ability to live independently
  - Aerobic bacterium become a mitochondrion
  - Photosynthetic（光合的）bacterium became a chloroplast
- rRNA and whole genome sequencing supports this hypothesis
  - Chloroplast DNA is very similar to cyanobacterial（蓝藻细菌的）DNA
  - Mitochondria DNA resembles that of *Rickettsia*（立克次氏体） bacteria



Phylogenetic tree

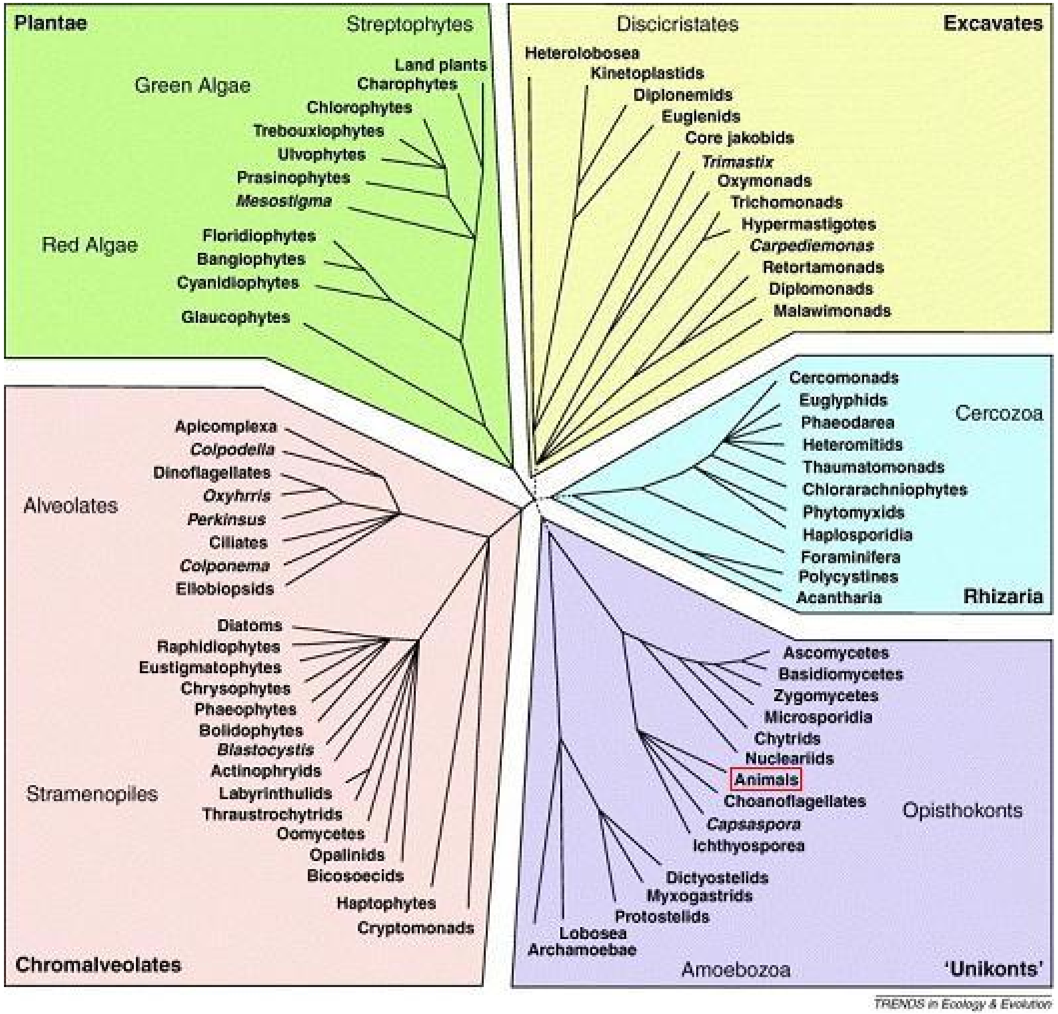
Introduction to Eukaryotic Microorganisms

- Animals and plants branch off near the tip of the phylogenetic tree
- A diverse array of eukaryotic microorganisms is known
- Major groups are:
  - Fungi
  - Protists（原生生物）
    - Protozoa（无脊椎原生动物）
    - Algae
    - Slime molds（粘菌类）

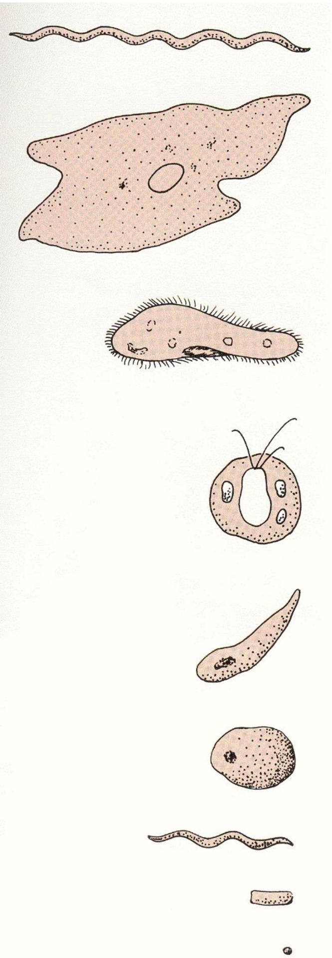


Phylogenetic tree of Eukaryotes still developing

- Tree is still being revised, based on sequences of 18S rRNA sequences as well as other genes and proteins
- Five supergroups are named
- Marine microbes appear in all of the five supergroups



Introduction to Eukaryotic Microorganisms: Diversity in size

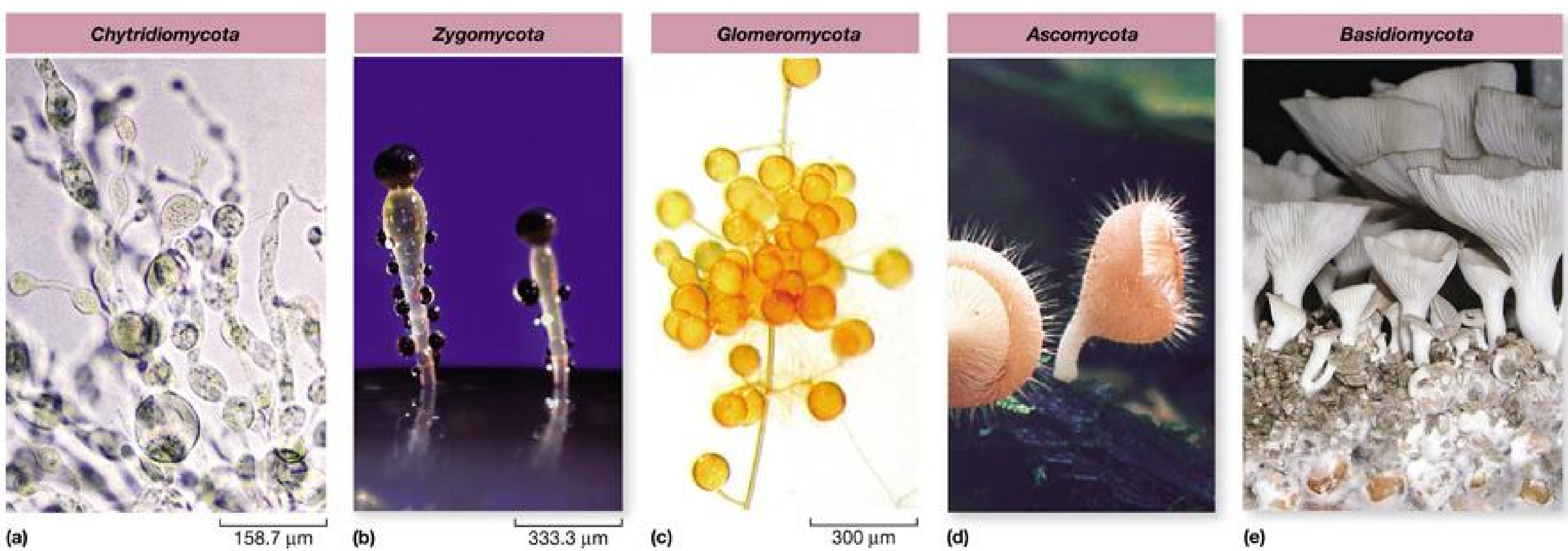


Genus and/or species or type	Size (length)	Grouping
Saprosphira（腐生螺旋属）	500μm	Bacteria
Giant amoeba（变形虫）	1mm	Protozoa
Paramecium（草履虫）	300μm	Protozoa
Chlamydomonas（衣藻）	25μm	Algae
Malaria parasite（疟原虫）	15μm	Protozoa
Yeast	10μm	True Fungi
Treponema pallidum（梅毒螺旋体）	10μm	Bacteria
Escherichia coli	3μm	Bacteria
Mycoplasma（支原体）	0.2μm	Bacteria

Although eukaryotes are generally larger than bacteria, the largest bacterium is bigger than most eukaryotes, and the smallest eukaryote is as small as some bacteria.

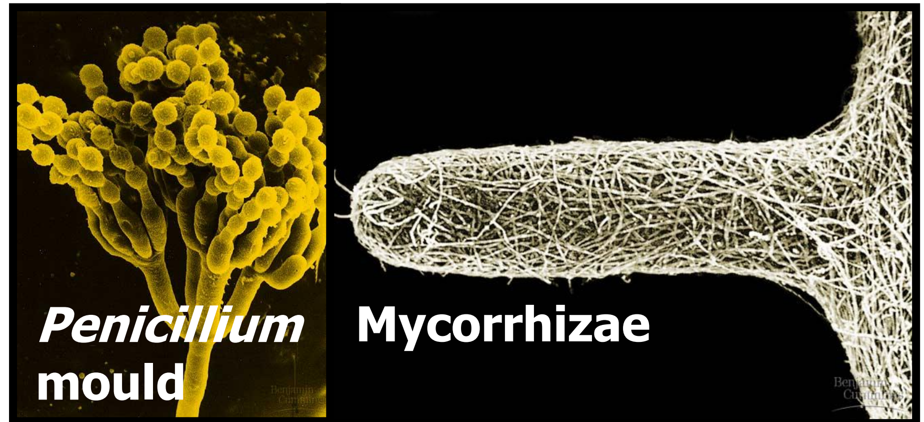
Introduction to Eukaryotic Microorganisms: Fungi

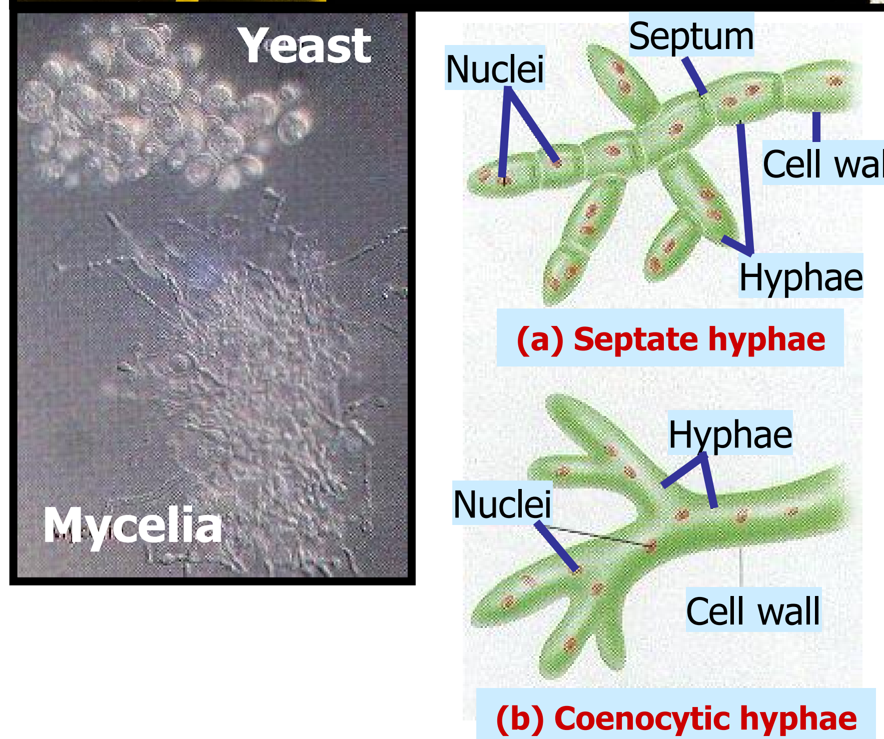
- **Fungus** (singular, s); **fungi** (plural, p)
- Fungi are sometimes called True Fungi, or Eumycota（真菌门）, or moulds
- **Mycology**（真菌学）is the study of fungi
- Fungi are eukaryotic microorganisms that:
  - Produce spores
  - Absorb nutrients
  - Lack chlorophyll(叶绿素)
  - Reproduce both sexually and asexually



Introduction to Eukaryotic Microorganisms: Fungi

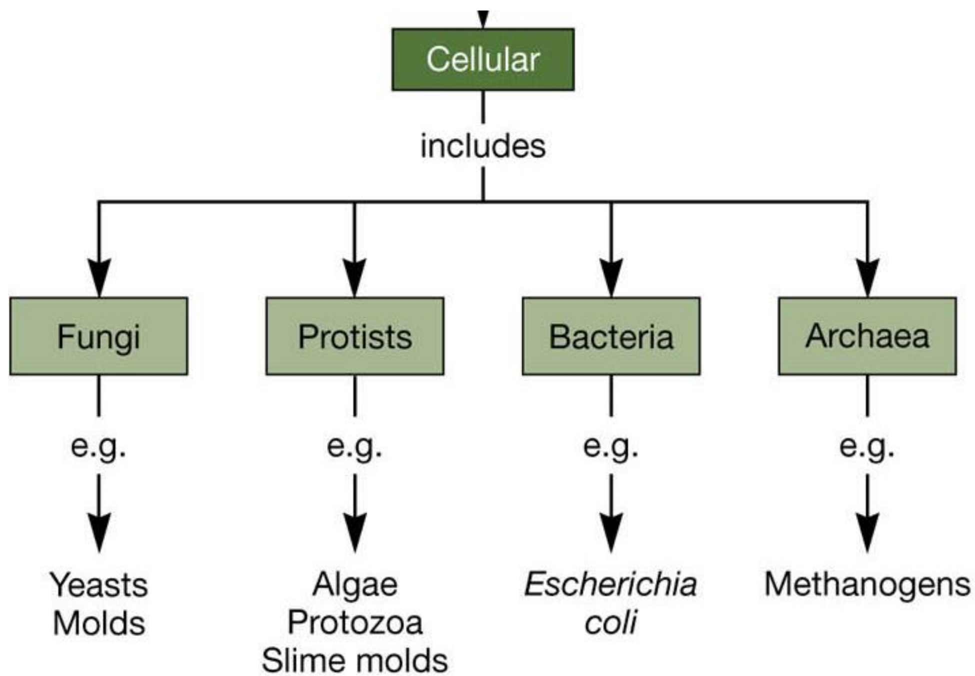
- Taxonomy is still being revised
  - Enormous（庞大的）group ~ 90,000 species described so far
- Distribution
  - Primarily terrestrial（陆生的）
  - Some are pathogenic（致病的）
  - Some are symbiotic（共生的）
  - Only ~0.05% are marine
- Structure
  - "body" is called a **thallus**（菌体）(varies from single yeast cells to multicellular moulds)
  - Thallus consists of long, branched filaments called **hyphae**(**菌丝**), tangled together into a **mycelium**(菌丝体)





Introduction to Eukaryotic Microorganisms: Protists（原生生物）

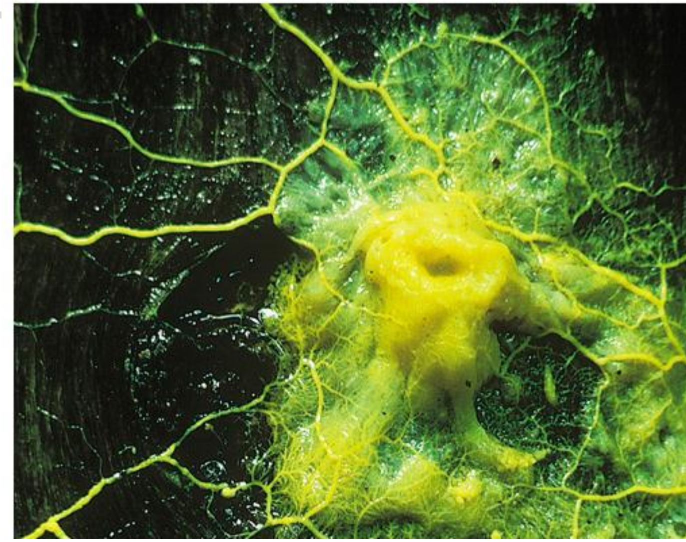
- Eukaryotic microorganisms have a confused taxonomic history
- The former Kingdom Protista（真核原生生物）is an artificial grouping not supported by modern phylogeny
- For convenience we refer to these organisms as Protists
- Protists do not have the tissue organisation found in fungi, plants and animals



- Protists include
  - **algae**
  - **protozoa**
  - **slime molds**

Introduction to Eukaryotic Microorganisms: Slime Molds

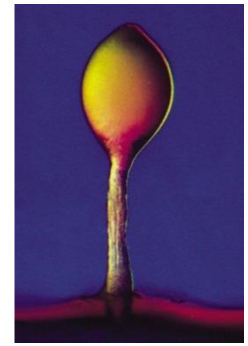
- Taxonomy: animals, plants or fungi??
  - Life cycles include motile (moving) amoeboid（似变形虫的） forms and fixed spore-forming body (fruiting body)
  - Acellular slime molds
  - Cellular slime molds
- Distribution
  - Terrestrial（陆生） and aquatic （水生） environments
- Nutrition
  - Heterotrophic（异养） saprophyte（腐生物） – feeds on organic matter



Acellular slime mold forming streaming mass of protoplasm which creeps along

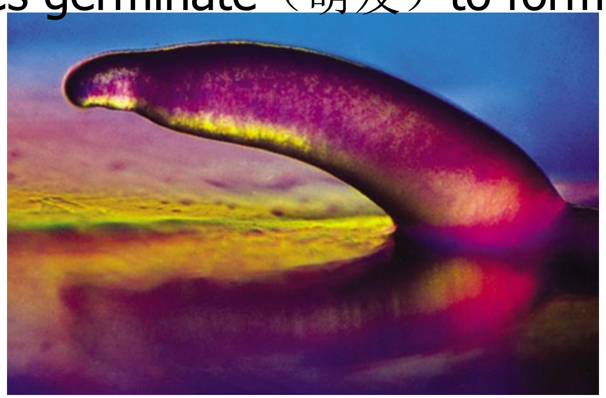
Acellular slime mold fruiting bodies

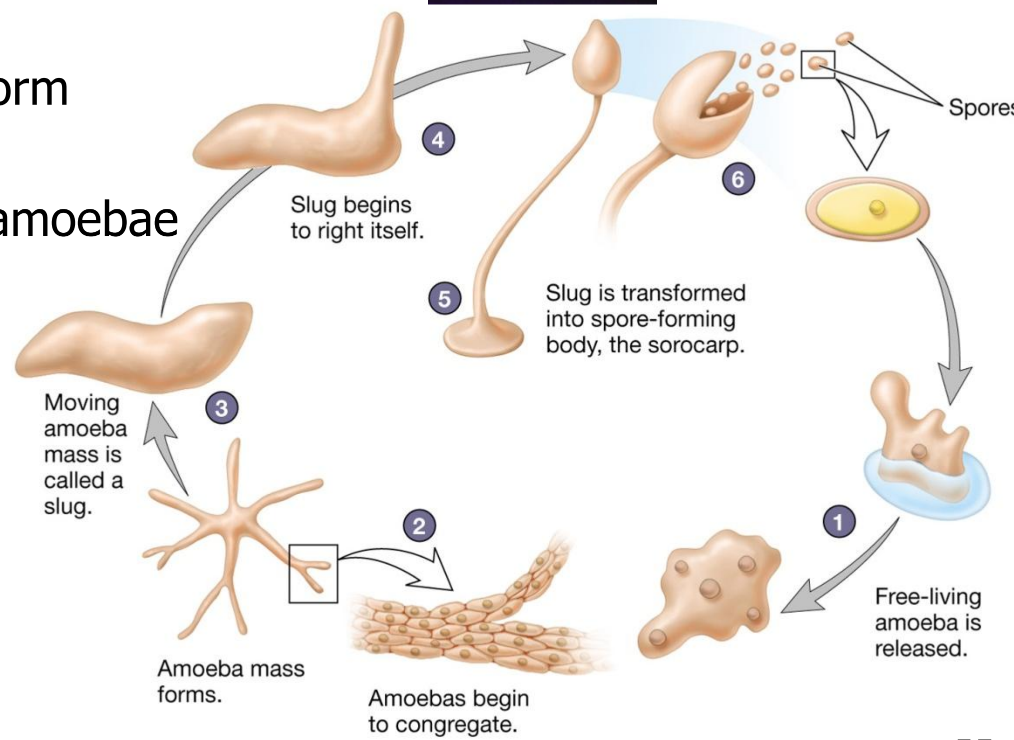
Introduction to Eukaryotic Microorganisms: Slime Molds



Life cycle for the Cellular Slime mould

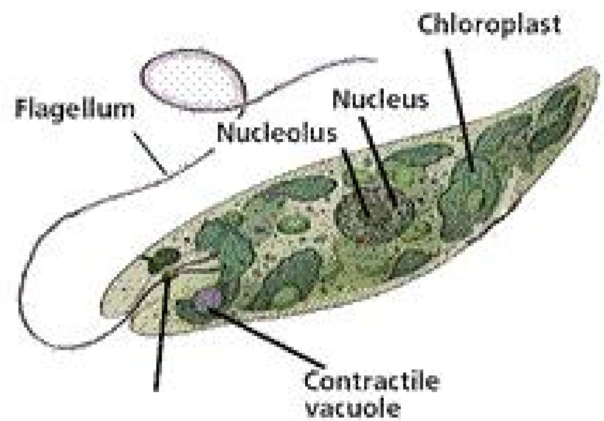
- Individual amoeba（变形）cells move together
- Form a multicellular slug which crawls（爬行）
- Slug（鼻涕虫）stops moving and form spores
- Spores germinate（萌发）to form amoebae





Introduction to Eukaryotic Microorganisms: Algae

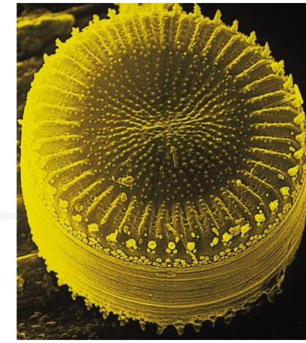
- Algae are photosynthetic（光合的） protists（原生生物）that (like plant cells) have a cell wall
- Microalgae: used to indicate unicellular algae (to distinguish them from larger, multicellular seaweeds)
- Algae are polyphyletic（多源的） i.e. they do not have a common ancestor, the groups are unrelated.
- Some important groups
  - Green algae
  - Red algae (seaweed 海藻)
  - Brown algae (kelp 巨藻)
  - Euglenids（褐藻纲）
  - Dinoflagellates（鞭毛藻）
  - Diatoms（硅藻）



Animal-like *Euglena*（眼虫属）

Green algae (*Spirotaenia*螺带鼓藻)

Introduction to Eukaryotic Microorganisms: Algae

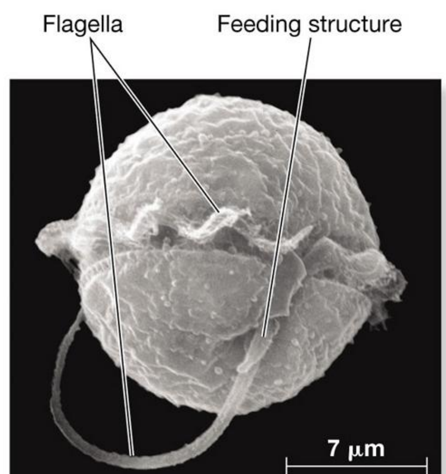


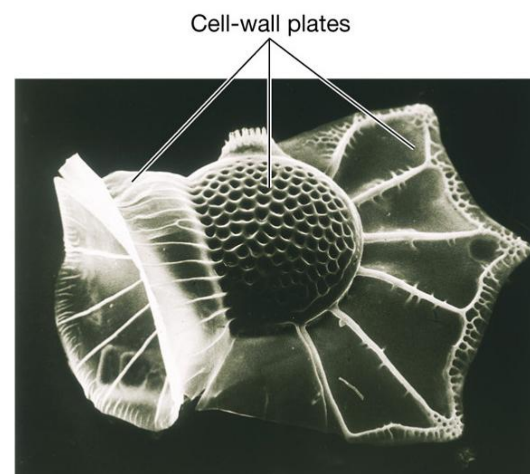
Diatoms（硅藻）

- Distribution
  - Widespread in the environment
    - Marine/Aquatic/Terrestrial（海生/水生/陆生）
    - Phytoplankton（浮游植物） (plankton = free-floating microbes in water environments)
    - Some cause algal blooms（水华）: sudden, rapid multiplication of plankton cells which may poison marine life forms
  - Some are endosymbiotic（内共生的） with:
    - Protozoans, Mollusks（软体动物）, Worms, Corals

Dinoflagellates（鞭毛藻）

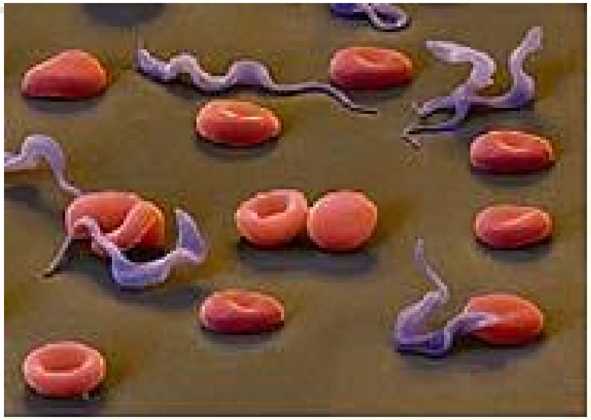
- Structure
  - Highly variable size/shape
  - Contain chloroplasts for photosynthesis（光合作用）
    - arose on several independent occasions during evolution of algae



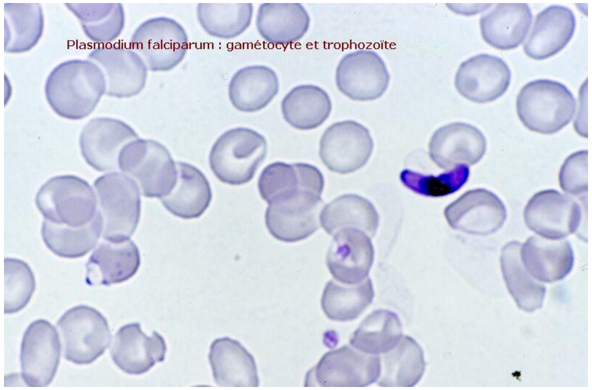


Introduction to Eukaryotic Microorganisms: Protozoa

- Historically defined as single-celled organisms with animal-like behaviors, such as motility（运动）and predation（捕食）
- Now used as a common term for a group of unrelated unicellular, non-photosynthetic protists
  - Chemoorganotrophic = use organic compounds（化合物）to gain energy, carbon and electrons
- Distribution
  - Marine/aquatic/terrestrial（海生/水生/陆生）
  - Some are animal pathogens（病原体）, e.g.
    - Sleeping sickness
    - Malaria (疟疾)
    - Amoebic dysentery (变形虫 痢疾)
    - Leishmania (锥体虫)



Trypanosoma brucei
(African Sleeping sickness)



Plasmodium falciparum
(Malaria)

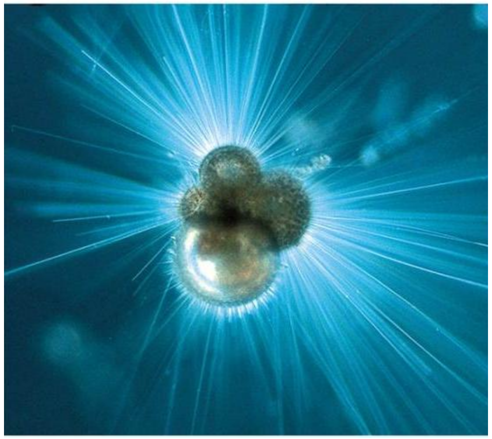
Introduction to Eukaryotic Microorganisms: Protozoa

- Structure
  - Flagellates: move with the help of whip（鞭子）-like structures called flagella（鞭毛）
  - Ciliates: move by using hair-like structures called cilia（纤毛）
  - Amoebae（变形虫）: move by the use of pseudopodia（伪足）– crawl（爬行）along surfaces by extending a pseudopod
  - Some do not move – sessile（固着的）

Telonema: a flagellate（鞭毛虫）

Acineria incurvata: a ciliate（纤毛虫）

- Some diatoms（硅藻）and some dinoflagellates（鞭毛藻）are protozoa (chemoorganotrophs, non-photosynthetic)



Formanifera（有孔虫）secrete shells of calcite（方解石）. They are amoeboid protozoa（无脊椎原生动物）.

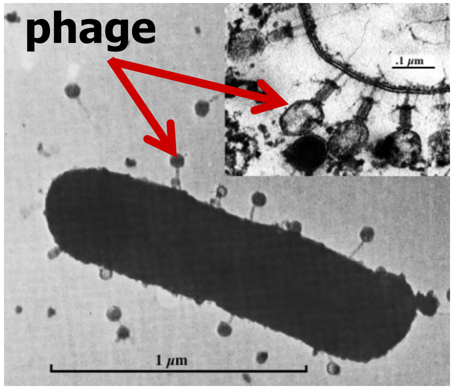
Tintinnids are ciliates which form vase-shaped cells

Introduction to Viruses

Distribution

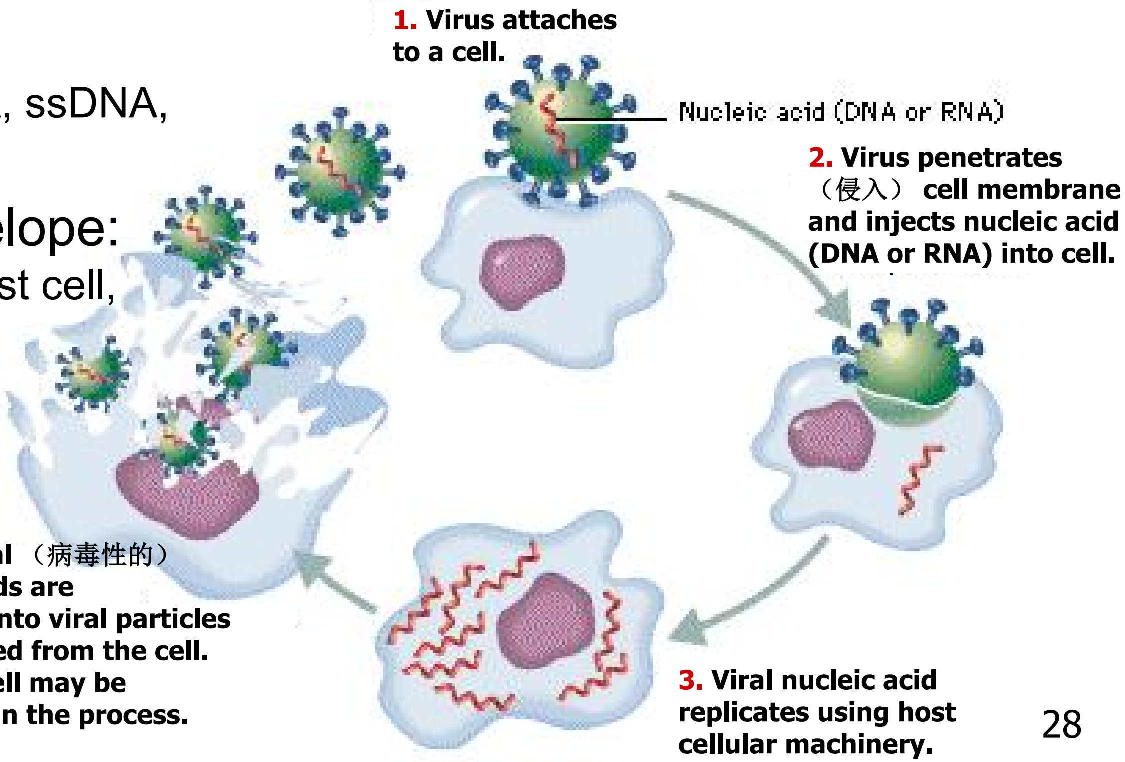
- Obligate intracellular parasites（寄生）of living cells
  - There are bacterial, plant, animal, and insect viruses

Bacteriophages (phages) infect bacteria



Taxonomy based on

- Nucleic acid type: dsDNA, ssDNA, dsRNA, ssRNA
- Presence/absence envelope: Lipids, carbohydrates from host cell, Proteins coded by virus
- Capsid（衣壳）shape



Introduction to Viruses: Structure

Structure

- Size 10-300 nm diameter
- Nucleocapsid core composed of DNA or RNA surrounded by a protein capsid
- Four morphological(形态的)types
  - Icosahedral(二十面体): 20 faces
  - Helical(螺旋型的): hollow (空心管) tube
  - Enveloped
  - Complex: e.g. with tail, like some phages

Bacteriophage Structure

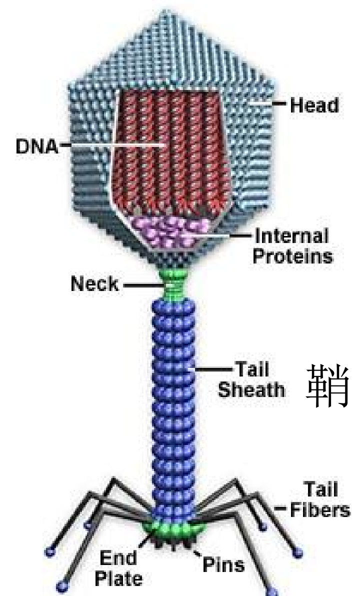


Figure 1

Animal Virus Structure

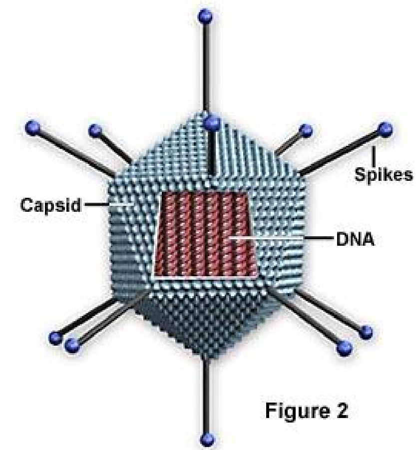
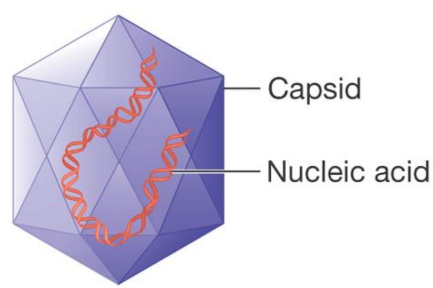
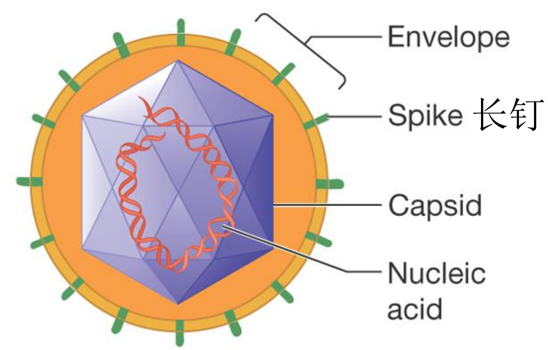


Figure 2

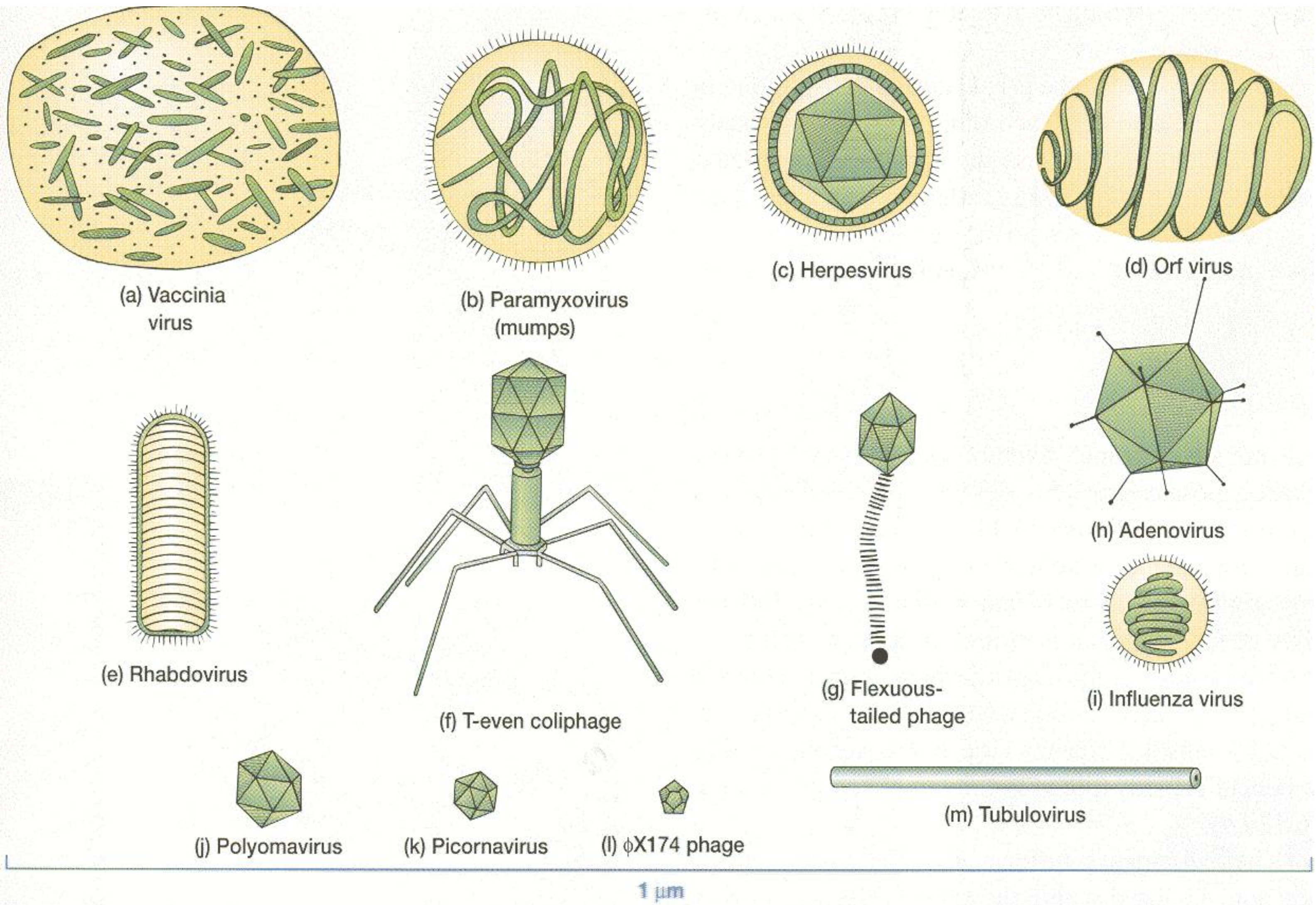


(a) Nonenveloped virus



(b) Enveloped virus

Introduction to Viruses: Size



a 牛痘病毒
b 副粘病毒
c 疱疹病毒
d 羊痘病毒
e 横纹肌溶解症病毒
f T型噬菌体
g 噬菌体
h
i 流感病毒
j 多瘤病毒
k 小核糖核酸病毒

References

Prescott's Microbiology 9th edition (PM). (some figures are taken from the 8th edition which may vary from 9th ed.)

Taxonomy and taxonomic ranks: chapter 19.1, 19.2
Diversity of microbes: 1
Domains of life: 1
Prokaryotes and eukaryotes: 1
Evolution: 1, 19.5
Introduction to fungi: 26.1
Introduction to protists: 25.1
Introduction to slime moulds: 25.3
Introduction to algae: 25.5, 25.6
Introduction to protozoa: 25
Introduction to viruses: 27.1

Objectives for this lecture:

At the end of this lecture, you should know the main structures of eubacterial and archaeal cells and their functions, including:

- Common cell shapes and arrangements
- Cell wall/peptidoglycan（肽聚糖）
- Plasma membrane (cell membrane/cytoplasmic细胞质的 membrane)
- Outer membrane and lipopolysaccharide（脂多糖）
- Periplasmic space（周质空间）
- Genetic material
- Flagella（鞭毛）
- Pili/fimbriae（菌毛）
- Endospores（内生孢子）
- Capsule（荚膜）, glycocalyx（多糖蛋白复合物
- ） and slime layers（黏液层）
- S layers

References:
Prescott's Microbiology 9th edition (PM)
Bacterial cell structure: PM chapter 3
Archaeal cell structure: PM chapter 4